

Yang Sen (Liam) Lin

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Education

National Yang Ming Chiao Tung University (NYCU) Sep 2021 – Dec 2025 (expected)
B.S., Mechanical Engineering & Computer Science (cross-disciplinary) GPA: **4.04/4.3** (last 60), 3.85/4.3 (overall)
Academic Excellence Award × 2 (top 5% of the dept.)
Ranking: **7/61 (11.4%)**

Relevant Coursework (Mathematics & Robotics): Engineering Mathematics (I) & (II), Automatic Control (I) & (II), Fluid Mechanics, Heat Transfer, Applied Mechanics (II), Computer Vision for UAV Autopilot.

Relevant Coursework (CS & AI): Intro to Algorithms, Data Structures and OOP, Computer Vision, Introduction to AI, Introduction to Image Processing, Computer Programming, Introduction to Database Systems.

Publications

[1] See, Point, Fly: A Learning-Free VLM Framework for Universal Unmanned Aerial Navigation. *CoRL 2025* 📄

Chih Yao Hu*, **Yang-Sen Lin***, Yuna Lee, Chih-Hai Su, Jie-Ying Lee, Shr-Ruei Tsai, Chin-Yang Lin, Kuan-Wen Chen, Tsung-Wei Ke, Yu-Lun Liu (* equal contribution)

(Note: CoRL ranked #3 in Robotics on Google Scholar, **H5-index: 107**)

Computational Impact: Proposed a learning-free framework that grounds 2D visual-language prompts into 3D metric controls via **geometric constraints**, achieving **SOTA zero-shot generalization** in unstructured environments.

Work & Research Experience

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Research Intern (Incoming) — Advisor: Dr. Masashi Hamaya

Tokyo, Japan

Jun. 2026 – Aug. 2026

- **Selected Research Topic:** Selected to develop a **knowledge distillation framework** to **compress large-scale Vision-Language-Action (VLA) models**, aiming to address **computational bottlenecks** for **real-time inference** on edge hardware.
- **Proposal Scope:** Plan to investigate **latency minimization techniques** for high-frequency control loops in **soft robotics**, with the goal of submitting findings to top-tier conferences (e.g., **CoRL/ICRA**).

Computational Photography Lab, NYCU

Undergraduate Researcher — Advisor: Prof. Yu-Lun Liu

Hsinchu, Taiwan

Sep. 2024 – Present

- **Geometric Formulation:** Derived a **mathematical action-to-control mapping** to ground high-level VLM inference (2D coordinates) into low-level drone dynamics (3D trajectories), **solving depth ambiguity** without depth sensors.
- **Sim2Real & Validation:** Overcame the sim-to-real gap by calibrating coordinate systems and **optimizing control latency**, validated through **55+ real-world flight experiments** demonstrating long-horizon planning.
- **Skills:** Python, VLM integration, Sim2Real, academic writing.

Wolley Inc.

Firmware Engineer Intern — Advisor: Dr. Yu-Ming Chang (Director)

Hsinchu, Taiwan

Jul. 2024 – Aug. 2024

- **System Profiling:** Engineered C-based diagnostic tools to profile **throughput and latency** for **CXL Type-3 solutions**, providing critical metrics to **reduce memory bandwidth pressure**.
- **Parallel Computing:** Refactored the internal testing system via **parallel computing techniques** in Linux, reducing unit testing time by **optimizing I/O efficiency**.
- **Skills:** C/C++, Linux, Virtual Machines, Parallelization, Algorithms.

Technical Skills

- **Languages:** Python (*Proficient*), C/C++ (*System-level & Embedded*), C# (*Unity*).
- **AI & Computer Vision:** OpenCV, YOLO, VLM, PyTorch, GANs.
- **Robotics & Control:** PID Control, UAV Dynamics, SDL (*Physics simulation*).
- **Systems & Tools:** Linux (*Profiling*), Git, Docker, **CXL/HDM Protocols**, Embedded Profiling Tools.

Selected Projects

Vision-Based UAV Autopilot System | *Python, OpenCV, Control Theory*

- **Algorithmic Optimization:** Optimized visual tracking algorithms from **quadratic $\mathcal{O}(N^2)$ to linear $\mathcal{O}(N)$ complexity** via **heuristic-based sparse sampling**, eliminating computational bottlenecks for real-time control.
- **Control Implementation:** Engineered a closed-loop **PID controller from scratch**, iteratively tuning proportional and derivative gains (K_p, K_d) to achieve **<3cm hover accuracy** without relying on external positioning systems.

Perceptual Image Restoration via High-Order Modeling | *Python, PyTorch, GANs*

- **Mathematical Modeling:** Modeled signal degradation via **Generalized Gaussian kernels** ($D = (k \otimes I)_{\downarrow s} + n$) to simulate complex blur artifacts, incorporating **high-order degradation processes**.
- **Perceptual Alignment:** Mitigated domain shift by fine-tuning generative priors, achieving superior **Mean Opinion Scores (MOS)** in perceptual fidelity compared to standard interpolation baselines.

DesignMate: Computational AR Sketching Interface | *Unity, C#, Linear Algebra*

- **Geometry Pipeline:** Developed a **vector-to-mesh pipeline** using **cubic interpolation** to smooth discrete stylus inputs, driving a diffusion-based generation workflow.
- **Resource Efficiency:** Designed a robust polling mechanism to manage **GPU memory allocation**, preventing bottlenecks on mobile AR hardware during distributed inference.

Leadership & Service

Google Developer Student Club (GDSC) NYC

Chapter Lead

Hsinchu, Taiwan

Jul. 2024 – Jun. 2025

- **Organizational Leadership:** Directed the university chapter comprising 24 staff across 5 departments; supervised **6 AI & Software Engineering project teams** to deliver technical solutions.
- **Impact:** Bridged the gap between academic theory and industry practice by organizing **10+ technical workshops** and project sharing sessions, impacting **40+ student developers** and fostering a peer-learning ecosystem.

Meichu Hackathon (Logitech Division)

Tech Lead & PM (3rd Place Winner)

Hsinchu, Taiwan

Sep. 2025

- **Technical Management:** Led a cross-functional team of 5 to develop *DesignMate* under a **48-hour deadline**; architected the system framework based on team expertise.
- **Communication:** Delivered the final technical pitch, demonstrating the ability to communicate complex engineering concepts to non-technical judges.

NYCU Varsity Swim Team

Team Member & Relay Gold Medalist

Hsinchu, Taiwan

Jan. 2022 – Dec. 2024

- **Achievement of Grit:** Progressed from a novice swimmer to a **City-Level Gold Medalist (Relay)** within 3 years. **Balanced 8+ hours of weekly high-intensity training** with rigorous research commitments.

Community Service

Homeless Support Volunteer

Taipei, Taiwan

Sep. 2025 – Nov. 2025

- **Social Initiative:** Initiated and executed a 10-week meal distribution program; managed procurement logistics and led weekly team reflection sessions to improve service delivery.

Honors & Awards

- **3rd Place**, Meichu Hackathon (Logitech Division) *Sep. 2025*
- **2nd Place & Popularity Award**, Taipei Metro Hackathon (*Ranked 2nd out of 92 teams*) *May 2024*
- **Gold Medal**, Men's 4x100m Relay, Hsinchu City Swimming Championship *Sep. 2024*
- **Academic Excellence Award** $\times 2$ (*Top 5% of Dept.*) *Fall 2023, Fall 2024*